

For this project, a dataset of 1000 images of 10 different breeds of ducks, with 100 images per breed, was created and then evaluated with the EfficientNetB0 model from Keras. The model was trained with 30 epochs and a batch size of 8 on an 80/20 train/test split.

The highest accuracy of 94.38% during training was at epoch 30, and the testing accuracy was 40.00%, as depicted in Figure 1. The very high training accuracy, paired with the relatively low testing accuracy, indicates that the model overfitted to the training data.

Future improvements that can be made are expanding the dataset to give the model more information/images to learn from. Further altering the training parameters may also result in a higher training accuracy.

Figure 1: Accuracy vs. Epoch

